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Water stress detection in potato plants using leaf temperature, emissivity, and reflectance

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ABSTRACT

Water stress is one of the most critical abiotic stressors limiting crop development. The main imaging and non-imaging remote sensing based techniques for the detection of plant stress (water stress and other types of stress) are thermography, visible (VIS), near- and shortwave infrared (NIR/SWIR) reflectance, and fluorescence. Just very recently, in addition to broadband thermography, narrowband (hyperspectral) thermal imaging has become available, which even facilitates the retrieval of spectral emissivity as an additional measure of plant stress. It is, however, still unclear at what stage plant stress is detectable with the various techniques.

During summer 2014 a water treatment experiment was run on 60 potato plants (*Solanum tuberosum* L. *Cilena*) with one half of the plants watered and the other half stressed. Crop response was measured using broadband and hyperspectral thermal cameras and a VNIR/SWIR spectrometer. Stomatal conductance was measured using a leaf porometer. Various measures and indices were computed and analysed for their sensitivity towards water stress (Crop Water Stress Index (CWSI), Moisture Stress Index (MSI), Photochemical Reflectance Index (PRI), and spectral emissivity, amongst others).

The results show that water stress as measured through stomatal conductance started on day 2 after watering was stopped. The fastest reacting, i.e., starting on day 7, indices were temperature based measures (e.g., CWSI) and NIR/SWIR reflectance based indices related to plant water content (e.g., MSI). Spectral emissivity reacted equally fast. Contrarily, visual indices (e.g., PRI) either did not respond at all or responded in an inconsistent manner.

This experiment shows that pre-visual water stress detection is feasible using indices depicting leaf temperature, leaf water content and spectral emissivity.

1. Introduction

Today, agriculture consumes about 80–90% of fresh water worldwide (Gonzalez-Dugo et al., 2010; Morison et al., 2008). Water-deficit stress, usually shortened to water- or drought-stress, is a phenotypic characteristic that exhibits dehydration in the plant due to the lack of available water required to keep cell concentrations at an acceptable and healthy level (Hopkins and Hüner, 2009). Water stress is one of the most critical abiotic stressors limiting plant growth, crop yield and quality concerning food production (Hsiao et al., 1976).

As a plant physiological reference, stomatal conductance measured by a porometer is the most sensitive reference measurement of plant water stress induced by water deficit (Jones, 2004a). A porometer measures the vapour concentrations at two different locations within a defined path using humidity sensors from which leaf transpiration is calculated. Knowing the leaf transpiration, stomatal conductance is directly calculated (Pearcy et al., 1989). However, the direct measure of leaf transpiration using porometry involves contact with the leaves and intervenes in the interactions between the leaf and the surrounding environment. Further, the method is labour-intensive, time-consuming, and only provides point measurements (Costa et al., 2013).

Imaging techniques open the possibility of fast and non-destructive spatio-temporal monitoring of many physiological and structural crop characteristics. The main remote sensing techniques for the detection of plant stress (water stress and other

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types of stress) are thermal imaging (thermography; 8–14 μm), visible, near- and shortwave infrared reflectance (VNIR/SWIR; 0.4–2.5 μm), and fluorescence (e.g., 0.76 μm). Thermal imaging takes advantage of the leaf energy balance equation, i.e. leaf temperature varies with transpiration from leaves (Jones, 1999; Tanner, 1963). The underlying concept is that a decrease in plant water status leads to a reduction in leaf transpiration as a result of active regulation of stomatal aperture, consequently increasing the leaf temperature due to a reduced evaporative cooling (Inoue et al., 1990). Conversely, a well-watered plant will have a lower temperature in comparison to the ambient air temperature. The stomatal regulation of a plant also highly depends on the actual meteorological factors, such as solar radiation, wind speed, ambient air temperature and vapour pressure deficit (VPD). VPD is the difference between the actual amount of water in the air and the maximum amount of water vapour in the air for a given temperature and therefore gives the capability of leaf transpiration. In general, the higher the VPD the higher the leaf transpiration rate for a healthy plant. In summary, leaf and canopy temperatures are a function of stomatal conductance and meteorological variables (Jones, 2004b). Stomatal closing is one of the first responses to water deficit and thus gives thermal imaging the possibility for detecting water stress pre-visual, i.e. before the change of leaf colour (Costa et al., 2013; Jackson et al., 1981; Jones, 2004b; Maes and Steppe, 2012).

The main problem of this temperature-based approach is constituted by the fact that the direct use of leaf or canopy temperature values alone cannot be an absolute estimator of the physiological status of crop plants (Inoue et al., 1990) since leaf temperatures measured under natural field conditions are very sensitive to highly fluctuating environmental factors. Thus, a variety of approaches were developed in the past to calibrate or normalize leaf temperature to estimate plant water stress more quantitatively. First corrections were done by the difference between leaf and air temperature (Idso et al., 1977). The popular Crop Water Stress Index (CWSI) (Jackson et al., 1981) not only corrects for air temperature but rather for all meteorological variables. In particular, the use of dry and wet artificial reference surfaces, which neglect additional meteorological data to describe the current environmental conditions, improved the usability of CWSI for thermal remote sensing (Jones, 1999).

Up to now, most studies regarding temperature based plant water stress detection used handheld broadband infrared cameras (Grant et al., 2006; Jones et al., 2009). The temperature retrieval using these 1 channel camera systems underlies the assumption of a constant emissivity (e.g. $\varepsilon = 0.97$ for vegetation), which in nature does not exist (Ullah et al., 2012). Thus, neglecting the spectral emissivity of the leaves themselves limits the accuracy of temperature estimation. For example, an error in the assumed emissivity of 1% results in absolute temperature errors of about 1 K (Jones, 2004b). Thermal hyperspectral imaging systems such as Telops HyperCam, Itres TASI-600, Specim AisaOWL have recently become available. The great advantage of these devices is the ability to measure the emitted radiance in many narrow bands, which compared to broadband thermal cameras allows a very stable temperature emissivity separation (TES) and very accurate temperature retrievals (Schlerf et al., 2012).

Besides temperature, these hyperspectral thermal infrared (TIR) imagers also have the ability to derive spectral emissivity by using a solid TES. Emissivity is defined as the ratio of radiative energy from the surface of an object of interest to the radiation from a blackbody following Planck's law depending on the wavelength at the same surface temperature. Until recently, emissivity spectra have not been exploited in vegetation studies for the following reasons: (1) low and complex spectral emissivity variations originate from plant physiological and biochemical processes as well as from plant structural effects; (2) low signal-to-noise ratio as well as low spatial and spectral resolution of airborne or satellite remote sensing TIR sensors fail to detect minor variations in plants' TIR spectral fingerprint; (3) to retrieve exact emissivity spectra accurate atmospheric correction and temperature emissivity separation (TES) methods are needed. Thus, only few authors studied plant properties in TIR mainly focusing on species discrimination (e.g., Ribeiro da Luz and Crowley, 2010, 2007; Salisbury, 1986; Ullah et al., 2012). Concerning plant stresses in the TIR, just very recently Buddenbaum et al. (2015) and Buitrago et al. (2016) demonstrated the ability of stress detection using the spectral emissivity. Buitrago et al. (2016) revealed the detection of water and cold stress at two different points in time based on the spectral emissivity of both European beech (*Fagus sylvatica*) and rhododendron (*Rhododendron cf. catawbiense*) leaves. They used a directional hemispherical reflectance (DHR) Bruker Vertex 70 FTIR laboratory spectrometer, equipped with an integrating sphere (Hecker et al., 2011). In comparison to passive emissive measurement devices, this non-imaging method is destructive, very time-consuming and limited to single leaves. On the contrary, Buddenbaum et al. (2015) were able to clearly differentiate water stressed and non-stressed European beeches (*Fagus sylvatica*) based on spectral emissivity using a passive emissive imaging spectrometer, the Telops HyperCam. No conclusions were drawn at which stage water stress can be detected using spectral emissivity in comparison to other stress sensitive methods (e.g., plant temperature, stomatal conductance), since this study was conducted during one diurnal course. Nevertheless, spectral emissivity facilitates new possibilities to detect water stress in contrast to the hyperspectral VNIR/SWIR spectral domain. Spectral radiation in the VNIR/SWIR is dominated by overtones and combination modes of fundamental vibrations, which originate from the interactions of solar radiation and leaf contents (e.g., leaf pigments). In comparison, spectral emissivity as the capability of emitting thermal radiation has large potential for the quantification of vegetation stresses, since it may be directly linked to leaf physiology and biochemistry (Ribeiro da Luz and Crowley, 2007).

Water stress not only changes leaf temperature and spectral emissivity but also leaf and canopy water content, pigment content, and structure. These leaf and canopy parameters are driving leaf and canopy reflectance in the solar reflective spectral range of the electromagnetic spectrum (VNIR/SWIR). Especially hyperspectral data opened the opportunity for the development of narrowband vegetation indices (VIs), which simplify the complex vegetation reflectance signatures and are indirectly related to plant physiological and structural parameters (Govender et al., 2009). Numerous studies make use of (VIs) for water stress detection such as the Water Index (WI, Peñuelas et al. (1997)), the Normalized Difference Vegetation Index (NDVI, Rouse et al. (1974)), and the Photochemical Reflectance Index (PRI, Gamon et al. (1992)). Suárez et al. (2009) observed robust relationships of PRI against canopy temperatures for various crops (e.g. $R^2 = 0.72$ for maize). Furthermore, Panigada et al. (2014) observed PRI as a more sensitive indicator for early plant water stress, when only plant physiological parameters are affected, than traditional VIs (e.g. NDVI). Additionally, VNIR fluorescence imaging is a good indirect estimator of water stress (Lang et al., 1996).

From the above discussion we derived the following aims for this study: (1) to evaluate in a controlled experiment the ability of temperature based indices and VNIR/SWIR reflectance based indices for detecting water stress in comparison to plant physiological measurements, (2) to compare hyperspectral and broadband imaging systems for deriving temperatures as a basis for water stress detection and (3) to examine the abilities of detecting water stress using the spectral emissivity of plant leaves.

Table 1
Number of observations per day after stress initiation.

Days after stress	0	1	2	7	8	9
HyperCam (emissivity)	59	59	58	56	60	60
HyperCam (temperature)	37	58	57	36	56	58
Fluke	60	10	48	59	60	60
ASD (leaf scale)	59	59	56	56	54	29
ASD (canopy scale)	60	59	59	–	15	60
Porometer	9	12	12	11	12	–
Weather station	60	59	59	59	60	60

2. Materials and methods

2.1. Experiment

In the period from July, 16th (0 days after stress) to 25th (9 days after stress) 2014, an outdoor water stress experiment on potato plants (*Solanum tuberosum* L. *Cilena*) was conducted under controlled conditions next to the greenhouse facilities of the Geobotany Department of Trier University (49°44′49.2″N, 6°41′02.4″E). The start of the experiment was set at the end of the flowering and the beginning of tuber initiation, when water consumption is highest (Dalla Costa et al., 1997). Potato tubers were sown in single pots (n = 60) with a substrate mixture of peat moss and sand (1:3, vol:vol). After the first day of measurement (16th July 2014) two treatments (control and treatment with n = 30 each), non- and fully-irrigated were applied. Plants were stored under a roof to prevent any external water supply (Fig. 1(b)). Before measuring, plants were moved outside to adapt to sunny conditions for at least half an hour. All data of the study were recorded during midday (11:00–15:30 CEST) under clear sky conditions. Bad weather conditions during the experiment prevented continuous measurements on a daily basis, which reduces the dataset to six dates (0, 1, 2, 7, 8 and 9 days after stress initiation). Furthermore, the number of observations (n) (Table 1) was mainly reduced due to failures of measurement devices and rapidly changing weather conditions. At the end of the experiment, tubers were harvested and the yields obtained for stressed and non-stressed plants were compared.

2.2. Meteorological measurements

The meteorological variables air temperature (T_{air}), air pressure, and relative humidity (RH), were measured at the study site using a portable automatic weather station (Vaisala WXT520) with a CR800 (Campbell scientific, Inc.) data logger. Measurements were recorded continuously at 10 s intervals at 2 m above ground. Following the findings of Anderson (1936),VPD is a more sensitive indicator of water vapour condition than relative humidity (RH) and therefore describes the interaction of plants with the intervening atmosphere more precisely. VPD (kPa) was calculated using Eq. (1) (Struthers et al., 2015):

$$\mathrm{VPD} = e_s \times \frac{100 - RH}{100} \times 0.1$$
$$e_s = 6.11 \times \exp\left(\frac{L}{Rv}\left(\frac{1}{273} - \frac{1}{T}\right)\right) \qquad (1)$$

where e_s is the saturation vapour pressure in mbar, L is the latent heat of vaporization (2.5×10^6 J kg^{-1}), Rv is the gas constant for water vapour (461 J K^{-1} kg^{-1}) and T is the actual air temperature (K). RH is the relative humidity (%).

2.3. Porometry

Plant physiological measurements of leaves, such as stomatal conductance (gs; μmol m^{-2} s^{-1}), leaf transpiration rate (E; mmol m^{-2} s^{-1}) and photosynthetic rate (A; μmol m^{-2} s^{-1}) were derived using a gas exchange porometer with a corresponding leaf chamber (LCi, ADC BioScientific Ltd., Hoddesdon, UK). Porometer measurements were carried out in the upper part of the canopy using only sunlit leaves at ambient illumination. Due to the self-calibration of the gas exchange system, the measurement takes approx. 10 min per leaf, limiting the number of observations to n = 12 plants per day. Measurements were taken automatically every 30 s. Leaves of the control and treatment group were measured alternately. To ensure that only measurements under clear sky conditions were taken into account, data points with a photosynthetically active radiation (PAR) greater than 400 μmol m^{-2} cm^{-1} were selected, which correspond well with the light saturation curve of potatoes (Pleijel et al., 2007).

2.4. Thermal infrared

2.4.1. Image acquisition

Hyperspectral TIR data were recorded using a HyperCam-LW (Telops Inc., Québec, Canada), which consists of an FTIR (Fourier transformed infrared) spectrometer using a 320 × 256 pixel MCT (mercury cadmium telluride) focal plane array (FPA) detector with up to 0.25 cm^{-1} spectral resolution, corresponding to 5 nm at 10 μm, in the spectral range of 7.7–11.5 μm (Schlerf et al., 2012). In simplified terms, this FTIR spectrometer works on the operating principles of a Michelson interferometer. It measures interferograms, which are translated into spectral radiances (W m^{-2} sr^{-1} cm^{-1}) using real-time discrete-Fourier transform. Further, the camera system is provided with two internal blackbodies with a known emissivity ($\varepsilon \geq 0.99$). The imaging spectrometer was equipped with a wide-angle telescope and a 45° tilted gold-coated mirror, which allowed vertical view with a field of view (FOV) of 672 mm × 538 mm and a pixel size of 2.1 mm at 1.5 m distance. A spectral bandwidth of 4 cm^{-1} FWHM (Full Width at Half Maximum) corresponding to 40 nm at 10 μm was chosen in the spectral domain from 869 to 1298 cm^{-1} (7.7–11.5 μm), resulting in 120 bands.

The broadband TIR images were taken by a Fluke TiR1 thermal imager (Fluke Inc., Everett, WA, USA), which has an uncooled microbolometer FPA of 160 × 120 pixel size and is sensitive in the spectral range of 7.5–14 μm. The emissivity coefficient is user selectable. Furthermore, the system is equipped with a co-registered 640 × 480 pixel visual camera, which facilitates a picture-in-picture image interpretation (Fluke Corporation, 2010).

Fig. 1 shows the experimental setup. Hyperspectral images were taken from a tripod with a sensor target distance of about 1.5 m. The Fluke TiR1 was mounted 1.2 m on top of the HyperCam-LW, due to its smaller FOV and in order to co-register both thermal imaging systems. Further, a highly diffused Infragold standard (Labsphere Inc, North Sutton, USA) was arranged within each scene for downwelling radiance (DWR) collection. Moreover, dry and wet artificial reference surfaces, leaf-shaped and consisting of a green felt, were placed in each image to determine lowest and highest possible leaf transpiration rates under actual micro-meteorological conditions. To improve the signal-to-noise ratio (SNR) of the hyperspectral data, eight consecutive images were recorded per plant. One measurement cycle took approximately four minutes. All 60 canopies were measured synchronously with the HyperCam-LW and the Fluke TiR1 in a consistent order for each day of the experiment.

2.4.2. Temperature retrieval

To compare the both TIR imaging systems, corresponding regions of interests (ROIs) were selected in each image pair (HyperCam and Fluke). Only sunlit leaves as well as sunlit parts of the dry and wet references were selected. A single ROI of a plant contained approximately 300 pixels for the Fluke TiR1 and 800 pixels for the

Fig. 1. Experimental setup with HyperCam on the tripod, Fluke TiR1, L*ci* leaf porometer, Infragold as well as dry and wet references targets (a), plant breeding and storage of the 60 potato plants under a roof construction (b), plants of the control (c) and treatment (d) group at day 9 after stress initiation.

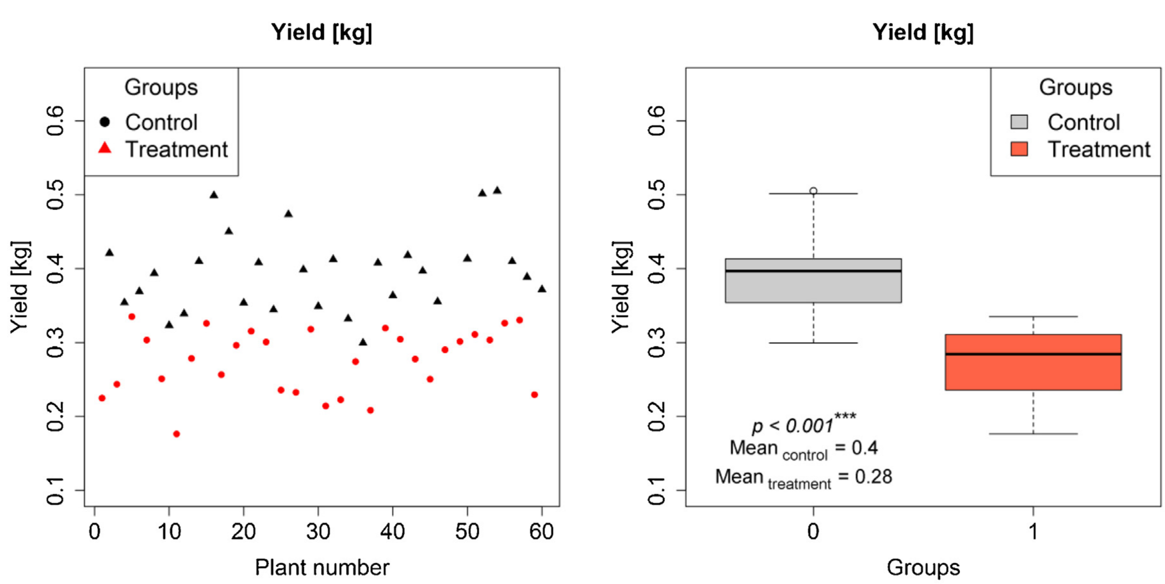

Fig. 2. Potato yield of stressed (treatment) and non-stressed (control) plants (left), boxplots and mean values per group as well as corresponding *p*-value of a *t*-test (right).

HyperCam-LW, due to its larger FOV. In an equal manner ROIs were drawn for the dry and wet references and the Infragold.

In order to avoid FOV-effects between the HyperCam and the Fluke, three temperature retrievals were performed depending on

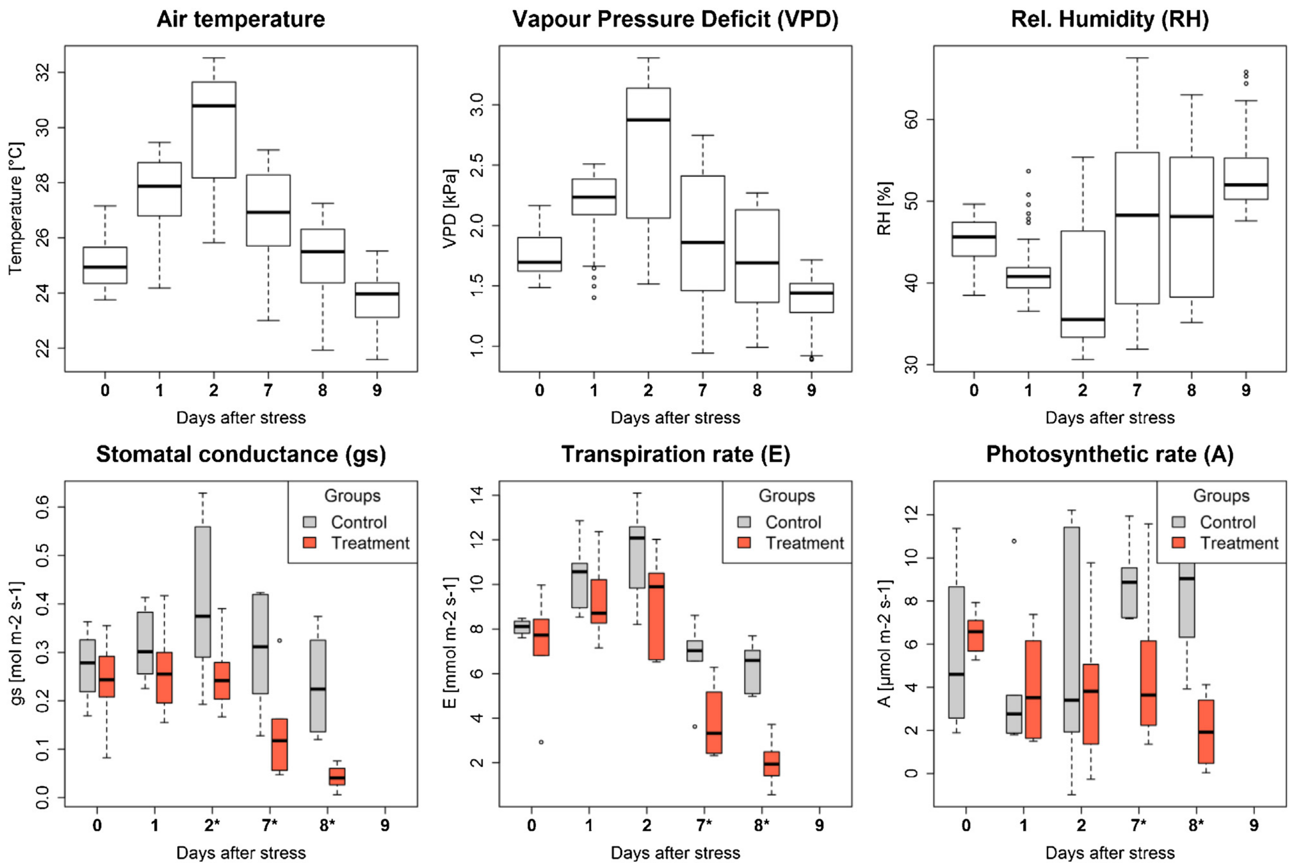

Fig. 3. Boxplots per day after stress initiation of the meteorological (air temperature, VPD, rel. humidity) and plant physiological measurements (stomatal conductance, transpiration and photosynthetic rate). Asterisks (*) indicating significant differences between control and treatment group at 5% significance level.

Table 2
Optical (VNIR/SWIR) narrow-band indices grouped by category: (1) xanthophyll pigments, (2) water content and (3) greenness. R is the reflectance at the ASD Fieldspec 3 wavelength in nm.

Category	Index	Equation	Reference
Xanthophyll	PRI	$PRI = (R_{570} - R_{531})/(R_{570} + R_{531})$	Gamon et al. (1992)
Greenness	NDVI	$NDVI = (R_{800} - R_{670})/(R_{800} + R_{670})$	Rouse et al. (1974)
Water content	WI	$WI = R_{900}/R_{970}$	Peñuelas et al. (1997)
	LWI	$LWI = R_{1300}/R_{1450}$	Seelig et al. (2008)
	MSI	$MSI = R_{1600}/R_{820}$	Hunt and Rock (1989)
	NDWI	$NDWI = (R_{857} - R_{1241})/(R_{857} + R_{1241})$	Gao (1996)

their origin: (1) 'spectral smoothness' TES based on hyperspectral radiances (HyperCam), (2) broadband temperatures based on hyperspectral radiances spectrally resampled to broadband radiance (HyperCam broadband) and (3) broadband temperature based on Fluke brightness temperatures (Fluke).

First, surface temperatures were retrieved from the hyperspectral HyperCam data. Therefore, the first step was to transfer the raw interferograms into spectral radiances ($W\,m^{-2}\,sr^{-1}\,cm^{-1}$). This radiometric calibration, including a Fourier transformation, a two-point calibration based on the internal blackbodies with known temperature and emissivity ($\varepsilon \geq 0.99$), as well as a bad pixel correction, was performed by Telops Reveal Calibrate software. Next, the so called 'spectral smoothness' approach (Horton et al., 1998) for temperature and emissivity separation (TES) was applied to retrieve surface temperatures from the spectral radiances. This state of the art TES approach allows for DWR correction and is most suited for outside measurements. The approach is only applicable for outside measurements under clear sky conditions and requires a short sensor-target distance to neglect distortions by the intervening atmosphere and to guarantee highly accurate surface temperature retrieval. It is based on the minimisation of isolated water vapour lines within the spectrum through changing the temperature. The temperature, which specified the smoothest spectrum, was chosen as surface temperature (T_s). We used the water absorption line at 1174.5 cm^{-1} (8.51 µm).

Second, concerning the possible FOV-effects between the two imaging systems a spectral resampling was applied on the hyperspectral HyperCam data. The resampling was done by summing up the spectral radiances per spectral band multiplied by the mean FWHM, resulting in a single spectral radiance over the spectral range from 869 to 1298 cm^{-1} (7.7–11.5 µm). Using a look up table (LUT) containing black body radiances with varying temperatures from 220 to 350 K at a resolution of 0.001 K, corresponding brightness temperatures were retrieved from the resampled spectral radiances. In comparison, the Fluke already provides images of brightness temperature. Lastly, surface temperatures of both broadband datasets (HyperCam broadband and Fluke) were calculated using Stefan-Boltzmann law corrected for DWR and an

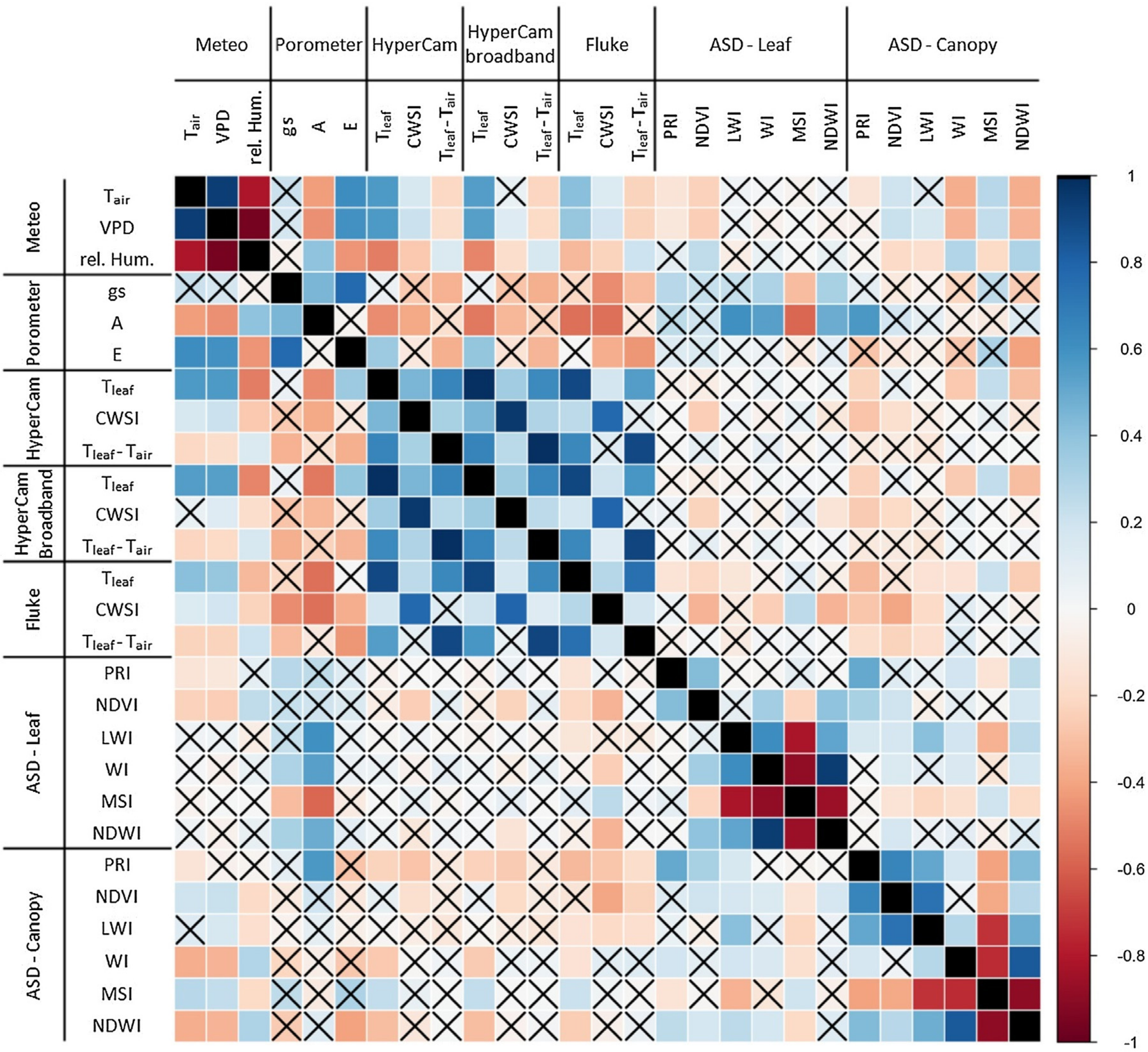

Fig. 4. Correlogram of spearman correlation of observed variables. Relationships which are not significant at 5% significance level are marked by crosses. Main correlation clusters occur within meteorological variables, comparing the three retrieval approaches of temperature based indices (e.g., CWSI) as well as water content based indices (e.g., WI and MSI). Furthermore, strong individual correlation can be observed, e.g. between T_{leaf} and meteorological variables.

adjusted emissivity for vegetation ($\varepsilon = 0.97$) (Maes and Steppe, 2012):

$$T_s = \sqrt[4]{\frac{T_{br}^4 - (1-\varepsilon) \times T_{bg}^4}{\varepsilon}} \quad (2),$$

Where T_S is the surface temperature of the object of interest, T_{br} is the brightness temperature measured by the sensor, and $(1-\varepsilon) \times {T_{bg}}^4$ is equivalent black body temperature for the DWR derived from the Infragold pixel within the specific image.

For further analysis we averaged the single temperatures per ROI, resulting in one temperature reading per potato plant and temperature retrieval approach.

2.4.3. Temperature based indices

Pure leaf temperatures can be used as a water stress detection estimator at one specific point in time, but are not suitable for water stress detection at several dates or during the course of the day, due to the interaction with rapidly changing meteorological conditions. To consider this fact, two temperature based indices were calculated: (1) the difference between leaf and air temperature ($T_{leaf} - T_{air}$) following the approach of the Stress Degree Day (SDD) (Jackson et al., 1977) and (2) the prominent Crop Water Stress Index (CWSI) using artificial wet and dry reference surfaces (Jones, 1999). CWSI was computed as follows:

$$CWSI = \frac{T_{leaf} - T_{wet}}{T_{dry} - T_{wet}} \quad (3),$$

where T_{leaf} is the leaf temperature, T_{wet} is the lower boundary for canopy temperature, assuming a leaf with stomata fully open and a potential transpiration rate of 100%, T_{dry} is the upper boundary compared to a non-transpiring leaf with stomata completely closed. Therefore, the use of artificial wet and dry references considers actual environmental conditions within the same image and no additional meteorological measurements are needed. In comparison, $T_{leaf} - T_{air}$ does correct for the fluctuation in T_{air}, but neglects the influence of VPD, solar radiation, or wind speed, which are considered in CWSI through the artificial reference surfaces.

2.4.4. Spectral emissivity retrieval

Knowing the DWR, which was measured using the Infragold standard, surface emissivity spectra are derived using Eq. (4):

$$\varepsilon_S(\sigma) = \frac{L_S(\sigma) - L_{DWR}(\sigma)}{L_{BB}(T_S, \sigma) - L_{DWR}(\sigma)} \quad (4)$$

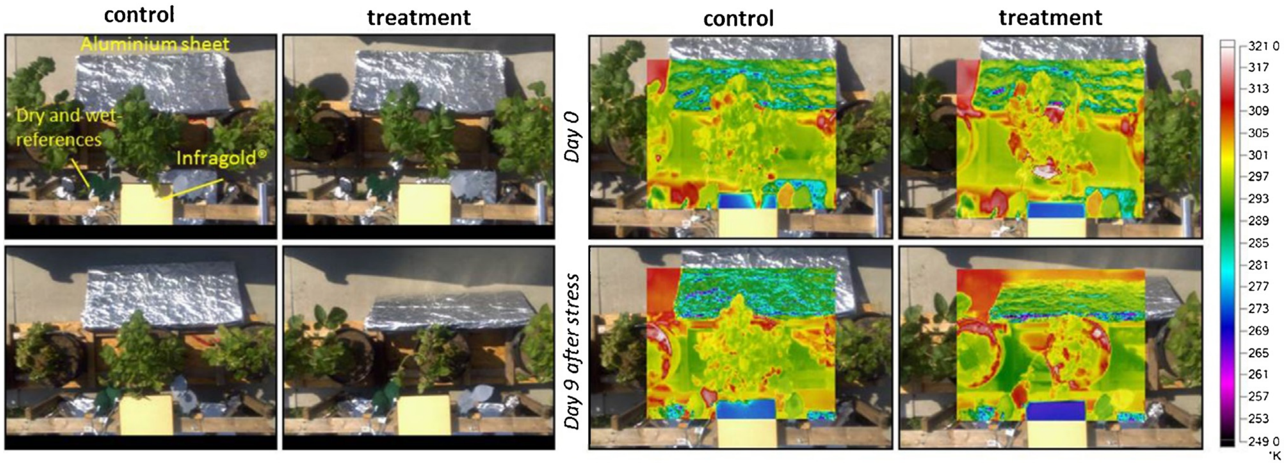

Fig. 5. Example of visual images (left) and corresponding temperature maps (right) taken with Fluke thermal imager at first and last day of the experiment. At the overlaid temperature maps Infragold reflecting the cold sky (blue coloured) at the bottom image edges can be registered as well as obvious temperature differences between the wet (yellowish green coloured with approx. 295 K) and dry reference targets (red coloured with approx. 310 K). Aluminium sheet was used to reduce the DWR from neighbouring objects. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where σ is the spectral wavenumber [cm^{-1}], e_s (σ) is the sample's spectral emissivity, L_{DWR} (σ) the downwelling radiance, and L_{BB} (T_s, σ) the blackbody radiance at the sample's temperature T_s. When using Eq. (4), mostly exact knowledge of the leaf kinetic temperature is required. This temperature (T_s) was retrieved using the 'spectral smoothness' approach (Horton et al., 1998).

Emissivity spectra were originated from the same ROIs as the HyperCam temperatures. For further analysis we averaged the spectral emissivities per ROI, resulting in one spectral emissivity reading per potato plant at several days of measurement.

2.5. VNIR/SWIR

To compare the sensitivity of water stress detection using TIR with common vegetation indices in the 0.35–2.5 µm spectral range, leaf as well as canopy spectra were measured using an ASD Fieldspec-3 spectrometer (ASD Inc., Boulder, CO, USA). The spectral resolution of the spectrometer is 3 nm at 0.7 µm and 10 nm at 1.4 µm and 2.1 µm. Leaf reflectance measurements were taken with an ASD Plant Probe, consisting of a clamp with a constant light source and turnable dark or white (Labsphere Spectralon reference) background with spot size of 10 mm. This setup ensures non-destructive leaf contact measurements under controlled illumination conditions. Canopy reflectances were measured with an ASD pistol grip under clear sky conditions. To account for specific illumination conditions a Spectralon reference panel was used to adjust the sensitivity of the detector. White reference as well as grating drift corrections were performed on each spectrum following Dorigo et al. (2006). For the further analysis we took an average spectrum for each plant, which consists of three leaves or parts of the canopy per plant times three single spectra per leaf times 30 scan repetitions per spectrum for both canopy and leaf reflectance measurements. Only leaves or parts from the upper sunny canopy were selected.

To examine the ability of detecting plant water stress using the VNIR/SWIR spectral region, several narrowband vegetation indices (Table 2) related to leaf physiology, structure and water content were calculated from the leaf and canopy reflectance spectra. The spectral bands centred at 0.531 µm and 0.57 µm were used to calculate Photochemical Reflectance Index (PRI) (Gamon et al., 1992). As three simple ratio indices related to water content, we estimated the Water Index (WI) (Peñuelas et al., 1993), the Leaf Water Index (LWI) (Seelig et al., 2008) and the Moisture Stress Index (MSI) (Hunt and Rock, 1989), which are sensitive in the scope of the water absorption bands at 0.97 µm and 1.45 µm respectively. Additionally, the Normalized Difference Water Index (NDWI) (Gao, 1996) and the Normalized Difference Vegetation Index (NDVI) were tested (Rouse et al., 1974).

2.6. Statistical analysis

To compare the three temperature retrieval approaches, a repeated measures ANOVA was used over all days of measurements. A repeated measures ANOVA considers the fact that the same samples were measured at multiple dates (days after stress initiation) and thus highly correlated. Further, statistical evaluation was performed on a daily basis to compare the sensitivity of the plant physiological measurements, the leaf temperatures and temperature based indices of the three temperature retrieval approaches and the vegetation indices regarding water stress detection. Graphical analysis was done by boxplots. Mann-Whitney-*U* tests at a level of significance of 5% were performed in order to examine significant differences between the two water treatments for the single days of the study. Furthermore, Mann-Whitney-*U* tests at a level of significance of 5% were also applied per wavenumber to analyse emissivity regarding differences in the spectra of control and treatment plants. Additionally, the relationships between all investigated variables are represented in a correlogram of Spearman's rank correlation coefficient. All analyses were performed in RStudio.

3. Results and discussion

3.1. Effect of water stress on yield

Application of water stress on potato plants had a highly significant effect ($p < 0.001^{***}$) on potato yield with average yields of 0.40 kg and 0.28 kg per plant for control and treatment, respectively. This confirms that during the experiment a severe water stress had been applied to the treated plants (Fig. 2).

3.2. Meteorological measurements

T_{air}, VPD and RH data were synchronised with the HyperCam measurements for the six study dates. T_{air} and VPD showed a very similar course over time (Fig. 3), increasing until day 2 and decreas-

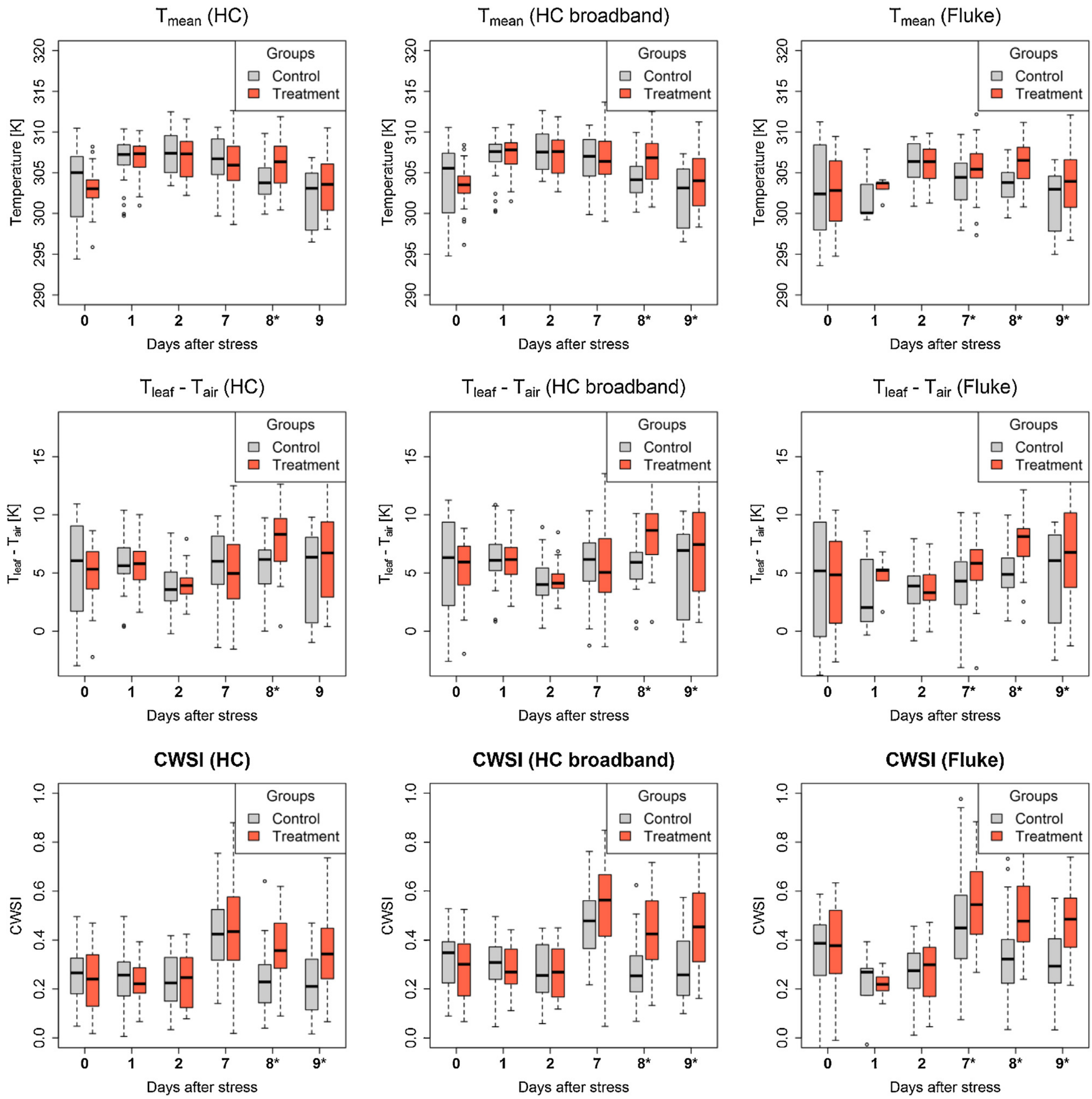

Fig. 6. Boxplots per day after stress initiation of leaf temperatures (T_{mean}), $T_{leaf} - T_{air}$ and CWSI of the three temperature retrieval approaches (HyperCam (HC), HyperCam broadband (HC broadband), Fluke). Asterisks (*) indicating significant difference between control and treatment group at 5% significance level.

ing afterwards until the end of the experiment. Mean T_{air} ranged from minimum 23.8 °C at day 9–29.9 °C at day 2 after stress initiation with an average over all days of 26.4 °C. On these same days VPD and RH minima (1.4 kPa and 40%) and maxima (2.6 kPa and 53%) also occurred. T_{air}, VPD and RH show strong correlations (Fig. 4). The meteorological variables reveal weather conditions suitable for the conducted water stress experiment.

3.3. Porometry

Stomatal conductance and transpiration rate show a high correlation ($r = 0.77^{***}$) (Fig. 4), whereas the relation between stomatal conductance and photosynthetic rate is weaker ($r = 0.45^{***}$). Boxplots of the three variables are shown in Fig. 3, grouped by treatment and control. Control and treatment groups are significantly different in stomatal conductance starting from 2 days after stress initiation ($p = 0.04^{*}$). The two groups are significantly different in transpiration rate and photosynthetic rate starting 7 days after stress (Table 3). Therefore, stomatal conductance is most sensible for water stress detection using plant physiological data as well as in comparison with TIR and VNIR/SWIR data. Further, the dependency of stomatal conductance and transpiration rate from meteorological variables (T_{air} and VPD) is well described by the courses of the boxplots; highest values can be found at day 2 where VPD and T_{air} are also at maximum for this study.

3.4. Thermal infrared

Fig. 5 shows an example of visual images and corresponding temperature images of plants from both water treatment groups at

Table 3
p-Values (level of statistical significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$) of Mann-Whitney-*U* test per day after stress initiation.

Device	Days after stress	0	1	2	7	8	9
Porometer	*n*	9	12	12	11	12	–
	gs	0.365	0.155	0.047*	0.041*	0.001**	–
	A	0.365	0.066	0.066	0.009**	0.001**	–
	E	0.365	0.066	0.066	0.009**	0.001**	–
HyperCam	*n*	37	58	57	36	56	58
	T_{leaf}	0.742	0.598	0.651	0.724	0.003**	0.063
	$T_{leaf} - T_{air}$	0.77	0.65	0.391	0.703	0.001**	0.057
	CWSI	0.658	0.785	0.554	0.365	<0.001***	0.002**
HyperCam broadband	*n*	37	58	57	36	56	58
	T_{leaf}	0.732	0.475	0.645	0.537	0.002**	0.043*
	$T_{leaf} - T_{air}$	0.722	0.586	0.367	0.512	<0.001***	0.046*
	CWSI	0.635	0.827	0.528	0.15	<0.001***	<0.001***
Fluke	*n*	60	#10	48	59	60	60
	T_{leaf}	0.607	0.155	0.569	0.035*	<0.001***	0.03*
	$T_{leaf} - T_{air}$	0.596	0.345	0.528	0.011*	<0.001***	0.036*
	CWSI	0.439	0.655	0.399	0.042*	<0.001***	<0.001***
ASD-Leaf	*n*	59	59	56	56	54	#29
	PRI	0.158	0.13	0.147	0.91	0.362	0.972
	NDVI	0.115	0.096	0.179	0.366	0.002**	0.177
	WI	0.165	0.118	0.097	0.024*	<0.001***	0.188
	LWI	0.461	0.291	0.293	0.237	<0.001***	0.2
	MSI	0.169	0.144	0.192	0.047*	<0.001***	0.035*
	NDWI	0.309	0.331	0.23	0.095	0.001**	0.234
ASD-Canopy	*n*	60	59	59	0	# 15	60
	PRI	0.354	0.479	0.101	–	0.047*	<0.001***
	NDVI	0.343	0.053	0.425	–	0.036	<0.001***
	WI	0.065	0.682	0.302	–	0.86	1
	LWI	0.195	0.21	0.345	–	0.06	<0.001***
	MSI	0.065	0.461	0.271	–	0.306	0.238
	NDWI	0.063	0.91	0.331	–	1	0.39

Reduced sample size.

the start and end dates of the experiment. The raw temperatures are not comparable due to rapidly changing meteorological conditions. However, from the visual images, structural canopy as well as colour differences between the first and the last day of measurement can be clearly recognized for the treatment, whereas the control group is still healthy.

3.4.1. Temperature comparison

To examine if a hyperspectral thermal imaging system provides more precise results concerning water stress detection based on temperatures in comparison to a broadband system, a first comparison of the three temperature retrieval approaches was conducted. The total mean temperature values for all days are very similar and differentiate only up to 0.6 K for the Fluke TiR1 and the HyperCam broadband temperatures and 0.2 K for the Fluke TiR1 and the HyperCam (hyperspectral). The difference between the two HyperCam approaches amounts to less than 0.4 K. This positive agreement is confirmed by a very high correlation coefficient $r = 0.99^{***}$ for the two HyperCam approaches (hyperspectral vs. broadband). Even the Fluke TiR1 is highly correlated with the HyperCam and HyperCam broadband temperatures with $r = 0.89^{***}$ and $r = 0.90^{***}$ respectively (Fig. 4). Moreover, the repeated measures ANOVA confirmed these findings showing no significant differences between the three introduced temperature retrieval approaches ($p = 0.12$).

Therefore, the choice of the TIR camera system had no significant effect on the retrieved temperatures. This finding implies that the assumption of $\varepsilon = 0.97$ for the broadband approaches fits excellently for the examined potato plants (*Solanum tuberosum* L. *Cilena*). A key advantage of a hyperspectral over a simple broadband imaging system is the possibility to correct for atmosphere effects. Not surprisingly, in this ground based experiment with a negligible amount of interferring atmosphere, this potential advantage played a lesser role. It is expected that hyperspectral systems outperform broadband systems in air- or spacborne missions, where high spectral resolution is essential for an appropiate atmospheric correction and TES like ARTEMIS (Automatic Retrieval of Temperature and Emissivity using Spectral Smoothness; Borel, 2003).

3.4.2. Water stress detection

Regarding the two HyperCam approaches, CWSI and $T_{leaf} - T_{air}$ are also highly correlated with $r = 0.96^{***}$ and $r = 0.99^{***}$ respectively (Fig. 4). The Fluke TiR1 shows also high correlations for the CWSI ($r = 0.78^{***}$ and $r = 0.79^{***}$) and for the $T_{leaf} - T_{air}$ ($r = 0.90^{***}$ and $r = 0.91^{***}$). The relationship of the T_{leaf} and meteorological data (T_{air} and VPD) is proved by correlation coefficients of up to 0.57^{***}. No statistically significant relationships can be found between simple leaf temperatures and stomatal conductance. Further, CWSI and difference between T_{leaf} and T_{air} are not very strong related to stomatal conductance. These small or non-existent relationships are mainly based on the small number of available observations for the plant physiological measurements.

For each variable, *p*-values between treatment and control group at every time point are listed in Table 3. Stomatal conductance (gs) shows significant differences between the two treatments of water supply starting from 2 days after stress initiation. At this point in time, none of the other variables respond. This is not surprising as gs is a genuine physiological variable and thus directly connected to the plant physiological response to stress.

Variables that respond at 7 days after stress comprise CWSI (computed from broadband TIR images), WI, and MSI based on leaf scale reflectance measurements in the NIR/SWIR spectral range.

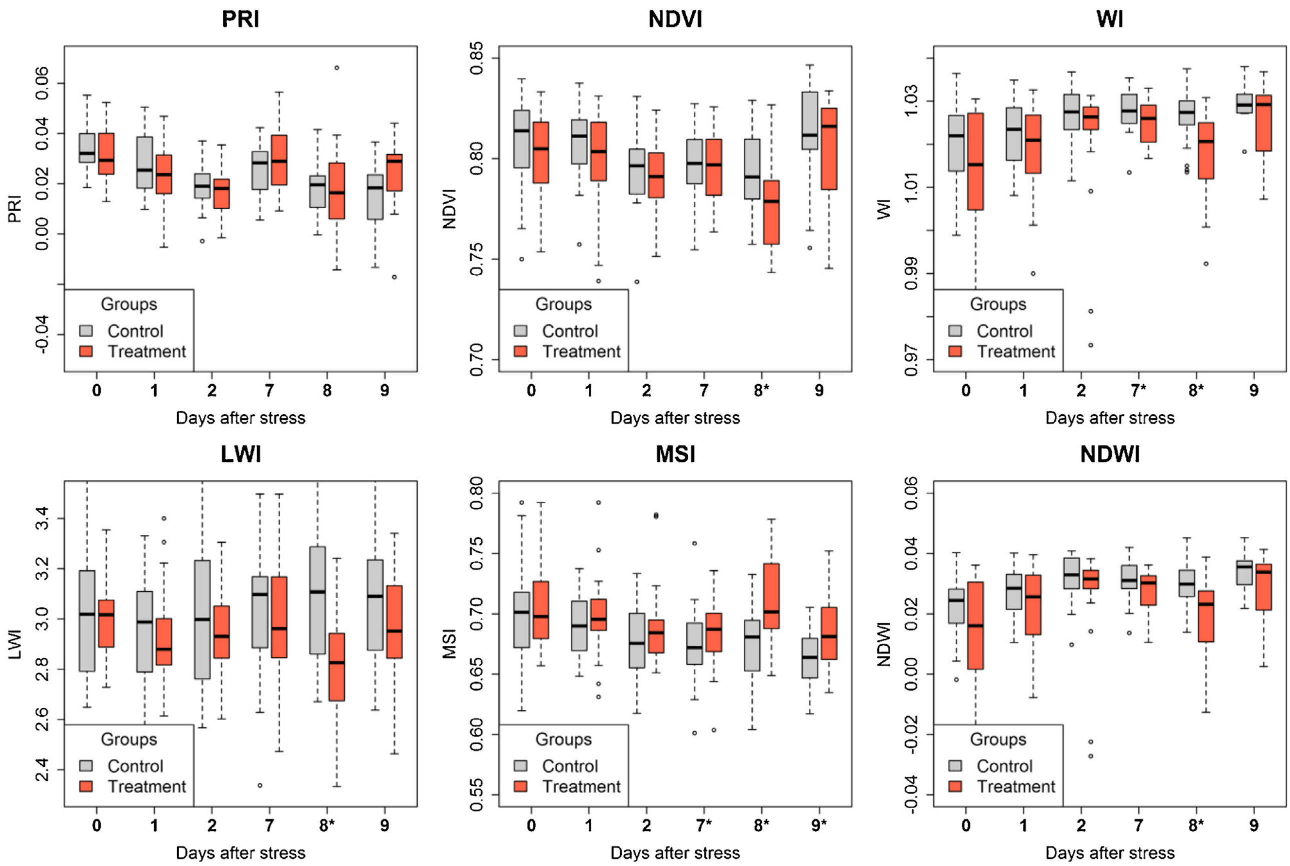

Fig. 7. Boxplots per day after stress initiation of vegetation indices at leaf scale. Asterisks (*) indicating significant difference between control and treatment group at 5% significance level.

Contrarily, indices in the VIS (PRI and NDVI) do not respond at all. Consequently, the responding variables are all of a pre-visual kind, i.e. not computed from wavebands located in the VIS domain.

Surprisingly, CWSI computed from HyperCam images only respond starting from 8 days after stress. Also the simple temperatures as well as $T_{leaf} - T_{air}$ show consistent results for the two broadband approaches (*$p < 0.05$), whereas the HyperCam temperatures ($p = 0.063$ and $p = 0.057$) are only significant at day 7 after stress initiation. Regarding the above mentioned temperature comparison these contradictory findings are mainly based on the very small differences in the corresponding p-values (i.e. for $T_{leaf} - T_{air}$ at 9 days after stress with $p = 0.057$ for the HyperCam and $p = 0.046^*$ for the HyperCam broadband), which is also obvious considering the boxplots (Fig. 6). These small differences, in addition to a relatively large number of observations (n = 58), last in statistically significant results at 5% significance level.

Therefore, CWSI is most suitable for water stress detection based on temperatures in comparison to the simple leaf temperature (T_{leaf}) and their correction with air temperature ($T_{leaf} - T_{air}$), due to its robustness against the actual meteorological environment. For example, decreasing of VPD (Fig. 3) from day 8 to day 9 leads to smaller absolute temperature differences between the two water treatment groups, which, comparing the three retrieval approaches, makes the results difficult to interpret. On the contrary, the CWSI shows consistent results for HyperCam, HyperCam-broadband as well as Fluke TiR 1 (Table 3).

A critical point of this study is the gap of measurements of four days during 19th and 23rd July 2014 due to unsuitable weather conditions. However, even if measurements had been taken during day 3–6 after stress initiation, we presume that CWSI would have not been sensitive to water stress before days 7–8 of the experiment. As a matter of fact, undisturbed outdoor experiments of this kind can hardly be guaranteed under the prevailing weather conditions in Central Europe.

3.5. VNIR/SWIR

All indices linked to water content are highly statistically significant (Table 3, Fig. 7) at day 8 after stress for leaf measurements. Only MSI showed consistent results starting from day 7 ($p = 0.047^*$), which is comparable to the results of the Fluke TiR1. Considering that water related indices are highly correlated, e.g. $r = 0.94^{***}$ for WI and NDWI, none of the observed indices seems to be more sensitive comparing the two water treatment groups. The relationship between water sensitive indices at leaf scale and stomatal conductance are weak, e.g. $p = 0.30^*$ for WI (Fig. 4).

NDVI and PRI are correlated with $r = 0.43^{***}$ for leaf and $r = 0.65^{***}$ for canopy scale. For the leaf measurements PRI showed no significantly differences between the two water treatments. Whereas, NDVI shows highly significant results ($p = 0.002^{**}$) at day 8 after stress initiation at leaf scale. On the contrary, PRI treatment and control were significantly different ($p = 0.047^*$) based on canopy measurements taken 8 days after stress. Similar results were found for NDVI. Thus, the findings of Panigada et al. (2014) that PRI is more sensitive to early stadium of plant water stress than NDVI could not be confirmed.

The results may be somewhat less robust due to the reduced sample size for leaf measurements at day 9 after stress (n = 29) and canopy measurements at day 7 (n = 0) and 8 (n = 15).

Overall, using leaf measurements and narrowband spectral vegetation indices linked to water content, water stress can be also detected at the same date as using CWSI. Therefore, TIR based

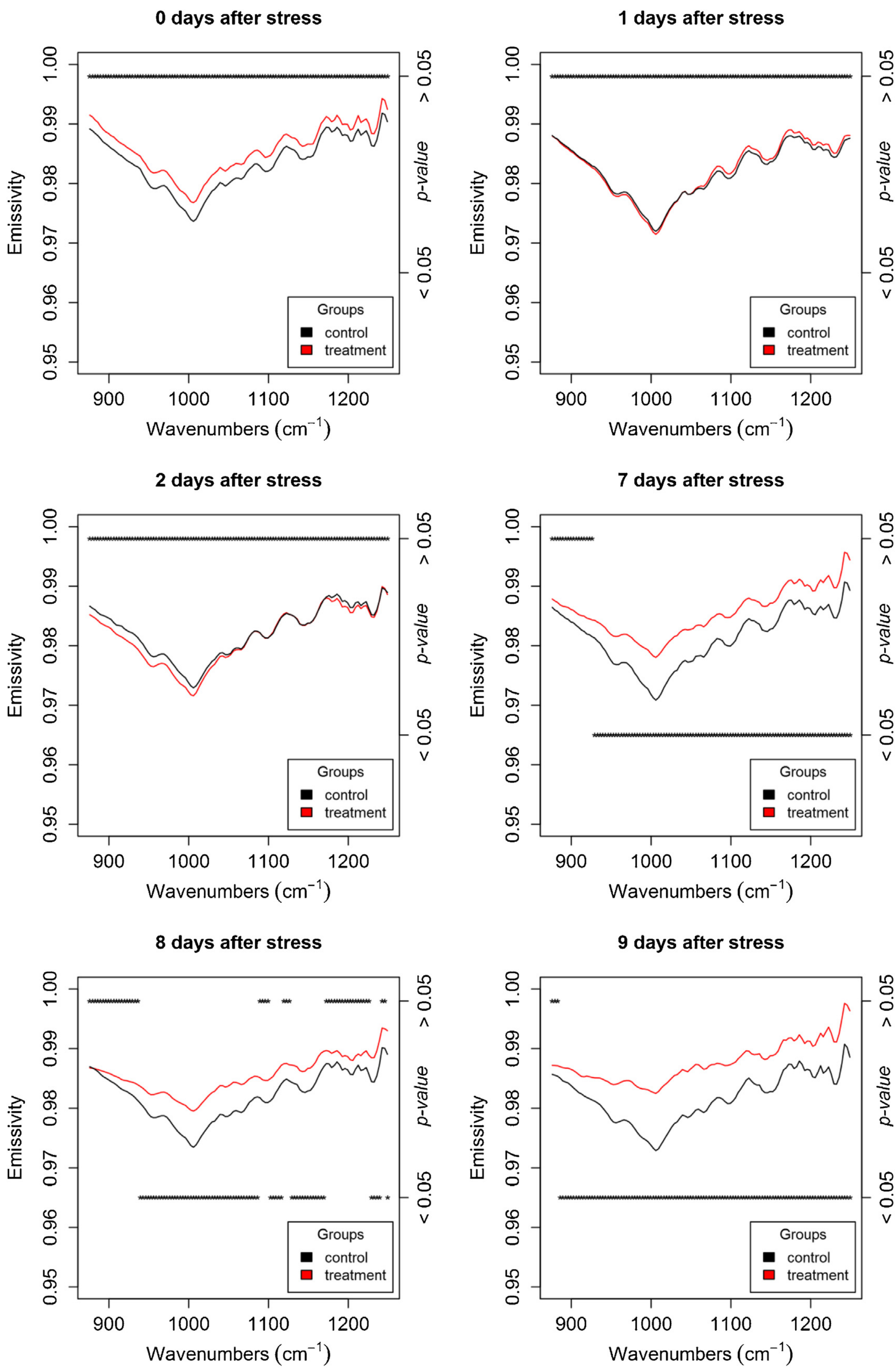

Fig. 8. Mean emissivity spectra of potato plants (*Solanum tuberosum L. Cilena*) per day after stress initiation for control (black) and treatment (red) group. Corresponding *p*-values of a Mann-Whitney-*U* tests at 5% level of significance per wavenumber are grouped by significance (right-handed and marked with asterisks. (For interpretation of the references to colour in this figure legend and text, the reader is referred to the web version of this article.)

indices and NIR/SWIR indices related to plant water content are equally sensitive and respond quicker than visual indices such as PRI and NDVI.

3.6. Spectral emissivity

Fig. 8 shows the mean emissivity spectra per day after stress initiation for control (black) and treatment (red) groups. All spectra shown are a consistent shape with a prominent spectral feature around 1000 cm^{-1} (10 μm). For example, weaker features were recognized around 950 cm^{-1} (10.53 μm), 1080 cm^{-1} (9.26 μm) or 1130 cm^{-1} (8.85 μm). For the first three days, the two water treatments were not significantly different (*p-values* > *0.05*; right-handed ordinate). However, control and treatment groups were significantly different (*p-values* < 0.05) starting from day 7 after stress initiation for a wavenumber greater than 925.99 cm^{-1} (10.79 μm) and even starting from 886.02 cm^{-1} (11.29 μm) for the last day of the experiment, respectively. At day 8 the differences were slightly less consistent, but still prominent. Therefore, spectral emissivity was equally sensitive to water stress as temperature based indices or NIR/SWIR based indices related to leaf water content.

For the water-stressed plants, emissivity increased in a consistent manner over all spectral bands, whereas the spectral shape remained unchanged. In particular, the spectral contrast around 1000 cm^{-1} decreased for the treatment group, while the control group constantly remained at an averaged emissivity value of 0.98 for all days of the experiment. Buitrago et al. (2016) examined changes in spectral emissivity for cold and water stress on two different species, European beech (*Fagus sylvatica*) and rhododendron (*Rhododendron cf. catawbiense*), respectively. They observed a similar emissivity increase for stressed plants of the rhododendron and related this increase to cavity effects (i.e., loss of spectral contrast due to multi scattering (Kirkland et al., 2002)) as a result of an increase of cuticle thickness. Similar results were also observed by Ribeiro da Luz and Crowley (2007). Regarding these findings, we expected that potatoes would react similarly to the beeches with a decrease in emissivity in the TIR rather than the rhododendron with its thicker and more complex cuticle compared to the examined potato plants.

Spectral features can be allocated neither to structural characteristics nor biochemical compounds, since corresponding reference measurements (e.g., thickness of cuticles or biochemical analysis) for the current study were not available. Notwithstanding, the overall increase in emissivity of the treatment group may most likely be related to cavity effects resulting from biochemical and structural changes in the leaf due to shortage of water.

Finally, we would like to highlight that this study resulted in a successful detection of plant emissivity changes with stress measured through an emissive imaging device. On the contrary, most other studies (e.g., Buitrago et al., 2016; Ribeiro da Luz and Crowley, 2007; Ullah et al., 2012) have based plant emissivity retrieval on non-imaging and slow reflectance-based (DHR) measurements. Therefore, such TIR imaging sensors have the potential to map plant emissivity from airborne platforms.

4. Conclusion

In this paper, we systematically compared temperature based measures, NIR/SWIR reflectance based indices and VIS reflectance based indices, as well as spectral emissivity in their ability to detect water stress in potato plants. Besides, we investigated hyperspectral and broadband TIR imaging devices for their ability to measure leaf temperature as the basis for the temperature based measures.

The main findings are: (1) water stress as measured through stomatal conductance started on the second day after watering was stopped; (2) temperature based indices and NIR/SWIR reflectance based indices related to plant water content as well as spectral emissivity were equally sensitive and responded quicker than visual indices such as PRI and NDVI; (3) spectral emissivity showed a steady overall increase with increasing water stress; (4) no striking differences appeared for hyperspectral and broadband TIR imagers in deriving accurate leaf temperatures; (5) among the temperature based measures, CWSI was most suitable for water stress detection, due to consideration of actual meteorological conditions based on dry and wet reference targets.

The most important conclusion of this research is that pre-visual measures of water stress (based on temperature, emissivity and water content) react faster than visual measures (based on leaf colour). The approach of pre-visual water stress detection is based on the physical nature of plant water reactions (i.e., stomatal closure due to water shortage). Thus, it should be possible to transfer this finding to other crop types. Under presence of water stress, pre-visual indices should also work at times in the growing season different to those that have been examined. Therefore, hyperspectral TIR measures like temperature based indices and especially spectral emissivity have an enormous potential for water stress detection and irrigation management. Nevertheless, following Costa et al. (2013) it should be pointed out that there is still a need for establishing thresholds for TIR based stress indices (e.g., CWSI) for different biotic or abiotic stress effects and for different species and varieties depending on the plant specific structure and stress strategies (e.g., hypo- or amphi-stomatous leaves).

The general validity of the experiment is slightly limited due to the four days gap of measurements and the reduced sample size of plant physiological and VNIR/SWIR reflectance measurements. Therefore, additional studies with comparable experimental setups should be conducted to confirm these results. Notwithstanding, given the typical weather conditions in Central Europe, it is in general very difficult to achieve over a sufficiently long period contiguous experimental datasets.

For potential application of TIR imaging in precision agriculture, the obtained results need to be confirmed at parcel and farm scales. The need for additional reference panel measurements for CWSI computation imposes a serious limitation for practical applications. Thus, there is a need to develop new approaches that characterise crop water stress independent from additional field measurements. Measures of spectral emissivity open new opportunities in TIR remote sensing and its application in agriculture. But, crop emissivity information is definitely limited by the availability of TIR imaging spectrometers, which are currently still very large in size and heavy (e.g., HyperCam with 30 kg) and thus, limited to operating on airborne platforms. Additionally, to fully exploit the emissivity signal, more fundamental research is needed to understand how spectral emissivity is affected by biochemical compounds, structural properties and the effects of stressors.

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